were named before telescopes were invented, so they could not yet be resolved into their stellar constituencies.

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As detailed in chapter 9, without the benefit of telescopes operating in multiple bands of light we might still declare the space between the galaxies to be empty. Aided by modern detectors, and modern theories, we have probed our cosmic countryside and revealed all manner of hard-to-detect things: dwarf galaxies, runaway stars, runaway stars that explode, million-degree X-ray-emitting gas, dark matter, faint blue galaxies, ubiquitous gas clouds, super-duper high-energy charged particles, and the mysterious quantum vacuum energy. With a list like that, one could argue that all the fun in the universe happens between the galaxies rather than within them.

In any reliably surveyed volume of space, dwarf galaxies outnumber large galaxies by more than ten to one. The first essay I ever wrote on the universe, in the early 1980s, was titled "The Galaxy and the Seven Dwarfs," referring to the Milky Way's diminutive nearby family. Since then, the tally of local dwarf galaxies has been counted in the dozens. While full-blooded galaxies contain hundreds of billions of stars, dwarf galaxies can have as few as a million, which renders them a hundred thousand times harder to detect. No wonder they are still being discovered in front of our noses.

Images of dwarf galaxies that no longer manufacture stars tend to look like tiny, boring smudges. Those dwarfs that do form stars are all irregularly shaped and, quite frankly, are a sorry-looking lot. Dwarf galaxies have three things working against their detection: They are small, and so are easily passed over when seductive spiral galaxies vie for your attention. They are dim, and so are missed in many surveys of galaxies that cut off below a prespecified brightness level. And they have a low density of stars within them, so they offer poor contrast above the glow of surrounding light from Earth's nighttime atmosphere and from other sources. All this is true. But since dwarfs far outnumber "normal" galaxies, perhaps our definition of what is normal needs revision.

You will find most (known) dwarf galaxies hanging out near bigger galaxies, in orbit around them like satellites. The two Magellanic Clouds are part of the Milky Way's dwarf family. But the lives of satellite galaxies can be quite hazardous. Most computer models of their orbits show a slow decay that ultimately results in the hapless dwarfs getting